

Ishita Rai

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EDUCATION

University of Pennsylvania - Master of Science in Computer Science

Relevant Coursework: AI, ML, CV, HCI, Data and Analysis for Marketing Decisions

May 2027

GPA: 3.90/4.0

University of Mumbai, India - Bachelor of Technology in Computer Engineering

Relevant Coursework: NLP, Algorithms, Deep Learning and Fuzzy Logic

Jun 2023

GPA: 3.92/4.0

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, TypeScript, C, SQL
- **Backend & Systems:** Flask, Spring, Node.js, REST APIs, GraphQL, Microservices
- **Frontend:** React, React Native, Tailwind, HTML/CSS
- **ML / Data:** PyTorch, TensorFlow, OpenCV, Spark
- **Developer Tools:** Git, Jenkins (CI/CD), Jira, Splunk, AWS, Docker

WORK EXPERIENCE

Software Engineer, JPMorgan Chase, Mumbai

Aug 2023 – Aug 2025

- Engineered and scaled a React/Flask/GraphQL production platform used by 500+ global users, reducing data-update latency from ~1 hour to minutes through backend workflow optimization
- Architected audit-safe workflows and notification pipelines for trade customization updates, ensuring 100% compliance, traceability, and reliable data lineage across downstream systems
- Migrated a legacy trade-update system to OpenFin-based React and Java micro-frontends, improving page load **performance by 96%** and achieving full user adoption
- Spearheaded modernization of a TypeScript UI codebase that lagged **10 major versions behind**, improving maintainability and reliability; presented the upgrade at the **Global Markets Technology UI Showcase**
- Deployed low-latency Java microservices, fault-tolerant APIs, and replay mechanisms for an FX electronic trading platform supporting 10K+ daily order events across 3 asset classes during peak market activity
- Designed a replay framework to recover lost downstream messages during transmission failures, preserving data consistency and preventing \$80K+ in potential trading losses

Software Engineering Intern, JPMorgan Chase, Mumbai

Jun 2022 – Jul 2022

- Deployed Flask-based microservice dependency visualization tool, reducing production root-cause analysis time by **80%**
- Built Hugging Face-powered FAQ chatbot, improving response accuracy by **~25%** via prompt engineering and retrieval tuning

Text Analytics Intern, KJ Somaiya College of Engineering, Mumbai

Feb 2020 – May 2021

- Engineered NLP preprocessing pipelines for 100K+ text samples using CoreNLP, NLTK, and spaCy, enabling large-scale information extraction and machine reading experiments
- Contributed pipeline design and experimental analysis to **IEEE-published research** on information extraction and predicate modeling

ACADEMIC PROJECTS

LLM-Based Adaptive Survey System (Python, OpenAI API, RAG)

- Built an LLM-based adaptive survey system using OpenAI API and RAG to identify ambiguity, staleness, and missing information in responses and generate targeted follow-up questions
- Designed multi-step dialogue workflows using missing attributes and model confidence to reduce redundant follow-up queries
- Created evaluation pipelines to compare completeness, ambiguity reduction, and coverage across response strategies, improving response completion by ~30%
- Automated transcript-to-structured-data pipelines for free-form responses for downstream analysis and LLM workflows

Community Detection in Large-Scale Heterogeneous Graphs (ML & Graph Mining)

- Benchmarked multiple graph-learning models and trained a Heterogeneous Graph Transformer for community detection in author networks, achieving 0.91 accuracy
- Prototyped a scalable graph-reduction pipeline that compressed a 40M-edge heterogeneous graph by ~50% while preserving semantic structure for efficient training

Kloris - PlantCare Mobile Application (React Native, Expo, Flask)

- Architected a full-stack plant-care mobile app featuring reminders, camera/GPS integration, and daily care workflows
- Implemented hybrid recommendation pipelines using clustering-based and content-based models with user profile and location/weather data to improve suggestion relevance

PennOS - Unix-like Operating System (C)

- Implemented Unix-like OS with concurrency control and a priority-based scheduler for preemptive multitasking

LEADERSHIP & VOLUNTEER EXPERIENCE

- **Research Assistant, Penn Medicine** (Incoming, Summer 2026): Working on safety guardrails for an LLM-based voicebot for dementia patient pre-screening in sensitive healthcare workflows
- **Teaching Assistant** (Spring 2026): Theory of Compiler Design, graded coursework, and mentored 75+ students
- **Penn FIFE Academy** (Fall 2025): Python Instructor for 30+ students, Grades 3–5, taught hands-on programming lessons
- **VP, Outreach: Pan-Asian GSA** (Fall 2025 – Present): Led outreach and community engagement for 1K+ members